



Endovascular thrombectomy for patients with large-core ischaemic stroke presenting up to 24 h after onset (ATLAS): a systematic review and individual patient data meta-analysis with central imaging adjudication

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Summary

Background Patients with extensive ischaemic change are often excluded from endovascular thrombectomy. We aimed to synthesise the evidence from recent trials in these patients by performing a systematic review and individual patient data meta-analysis to estimate treatment benefit, including within clinical and imaging subgroups.

Methods In this systematic review and meta-analysis, we searched PubMed and Embase for randomised trials published between March 1, 2018, and March 1, 2025, that evaluated efficacy and safety of endovascular thrombectomy compared with medical management in patients with large-core ischaemic stroke (based on an Alberta Stroke Program Early CT Score [ASPECTS] of ≤ 5 or estimated ischaemic core ≥ 50 mL) presenting within 24 h of onset. Individual patient-level data from all eligible trials were obtained. A central imaging core laboratory readjudicated ASPECTS and reanalysed ischaemic core volume. A two-stage meta-analysis with random-effects model was used to evaluate the distribution of 90-day modified Rankin Scale (mRS) scores (the primary outcome) using adjusted pooled generalised odds ratios (aGenORs). Missing data were handled by multiple imputation. Safety outcomes were all-cause mortality within 90-day follow-up and neurological worsening within 24–48 h of randomisation, reported as adjusted pooled relative risk (aRR); and symptomatic intracerebral haemorrhage within 36 h of randomisation (reported as risk difference). Subgroup analyses based on clinical and imaging characteristics were done, including subgroups defined by ischaemic core volume, ASPECTS, and time window from onset to randomisation. The meta-analysis was registered with PROSPERO (CRD420251058584).

Findings We included 1886 patients (944 assigned to endovascular thrombectomy and 942 assigned to medical management) from six trials. Baseline characteristics were similar between treatment groups. At day 90, the distribution of mRS scores was improved in patients in the endovascular thrombectomy group (median score 4 [IQR 3–6]; $n=940$) versus those in the medical management group (5 [4–6]; $n=931$; aGenOR 1.63 [95% CI 1.42–1.88], $p<0.0001$). The endovascular thrombectomy group also had reduced mortality (292 [31.1%]) compared with the medical management group (347 [37.3%]; aRR 0.82 [95% CI 0.70–0.97], $p=0.022$). No significant differences were observed in symptomatic intracranial haemorrhage (ten [1.1%] of 944 vs nine [1.0%] of 942 patients; pooled unadjusted risk difference -0.17 percentage points [95% CI -1.01 to 0.67], $p=0.69$) or neurological worsening (197 [22.0%] of 896 patients vs 161 [17.9%] of 899; aRR 1.19 [0.87–1.62], $p=0.27$). Improved functional outcomes with endovascular thrombectomy were consistent across clinical and imaging subgroups, except for those with an estimated ischaemic core volume of 150 mL or greater, in whom point estimates favoured endovascular thrombectomy, particularly in the early time window (0–6 h), but wide 95% CIs limited interpretation.

Interpretation Endovascular thrombectomy was associated with improved functional outcomes and reduced mortality versus medical management in patients with large-core ischaemic stroke presenting within 24 h of onset. With the exception of very extensive ischaemic changes (core volume ≥ 150 mL) presenting beyond 6 h, where evidence remains limited, benefit was sustained across ASPECTS and ischaemic core strata for patients presenting up to 24 h after onset.

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Introduction

The pivotal trials that established the efficacy and safety of endovascular thrombectomy for ischaemic stroke due to large vessel occlusion preferentially recruited patients with small ischaemic core, and largely excluded patients with extensive ischaemic injury, who comprise approximately 20% of those with large-vessel occlusion.¹ Thus, evidence supporting use of endovascular thrombectomy in patients with extensive ischaemic strokes has been limited, and these patients were often excluded from receiving endovascular thrombectomy. Since then, multiple randomised trials have shown improved outcomes with endovascular thrombectomy along with reasonable safety profiles in patients with low Alberta Stroke Program Early CT Score (ASPECTS; ≤ 5) or large ischaemic core volume. These trials reported consistent results, with five trials reporting a positive primary outcome^{2–6} and the sixth remaining directionally favourable (posterior probability of benefit 95·6%).⁷

These trials used different imaging modalities and eligibility criteria to identify patients with large

ischaemic core (appendix pp 27–28). An ASPECTS of 5 or lower (based on non-contrast CT or MR diffusion) or an ischaemic core volume of 50 mL or greater (based on CT perfusion or MR diffusion) were used in various combinations in 0–7, 0–12, or 0–24 h time windows after stroke onset. Outcome measures and overall treatment effect estimates for endovascular thrombectomy also varied between these trials. In addition, the precision of the effect estimates in key subgroups was limited by the small numbers of individuals in these subgroups enrolled in individual trials, resulting in persistent uncertainty regarding the benefit of endovascular thrombectomy in patient subgroups such as those with extremely large ischaemic core, those with absence of perfusion mismatch, and older patients.

We did a systematic review and individual patient data meta-analysis of trials evaluating endovascular thrombectomy in patients with low ASPECTS or large ischaemic core to improve understanding of the overall benefits and risks of endovascular thrombectomy in this population and to improve the precision of treatment

Research in context

Evidence before this study

Endovascular thrombectomy reduces disability for selected patients with acute ischaemic stroke due to large-vessel occlusion in the anterior circulation who present within 24 h of symptom onset, as shown in previous individual patient data meta-analyses of randomised trials. However, patients with extensive ischaemic injury, who comprise approximately 20% of those with large-vessel occlusion, were often excluded from the trials included in those studies and have not been routinely treated in clinical practice. Recent trials have shown efficacy and safety of thrombectomy specifically in patients with extensive ischaemic injury. In this systematic review and individual patient-data meta-analysis, we searched PubMed, Embase, and the ClinicalTrials.gov registry using keywords “stroke”, “thrombectomy”, “large core”, “ASPECTS”, and “ischaemic core”, with results limited to human clinical trials published between March 1, 2018, and March 1, 2025. This individual patient data meta-analysis aims to improve the precision of treatment effect estimates, including in important clinical and radiological subgroups.

Added value of this study

This systematic review and individual patient data meta-analysis of six randomised trials confirms an overall functional benefit and reduced mortality with endovascular thrombectomy in patients with a large-core ischaemic stroke and improves the precision of effect estimates in key subgroups of patients compared with those available in individual trials. The magnitude of reduction in disability was clinically meaningful, with numbers-needed-to-treat of 4·2 (95% CI 3·3–5·8) to improve at

least one modified Rankin Scale point, 8·6 (6·2–14·1) for improved functional independence, 5·7 (4·7–7·2) for independent ambulation, and 14·6 (7·5–326·3) for reduced mortality at 90-day follow-up. No significant differences in symptomatic intracranial haemorrhage or early neurological deterioration were observed. The primary functional outcome showed no effect modification by age, clinical stroke severity, site of arterial occlusion, or affected hemisphere. Patients benefited regardless of Alberta Stroke Program Early CT Score (ASPECTS) or perfusion imaging mismatch. Endovascular thrombectomy was beneficial for patients with an ischaemic core volume up to 150 mL in the 0–24 h time window; evidence was limited above this volume, with wide 95% CIs, although point estimates remained favourable, overall and at 0–6 h after onset.

Implications of all the available evidence

These results confirm the efficacy and safety of endovascular thrombectomy in a substantial group of patients with extensive ischaemic injury who were previously not considered eligible for thrombectomy. Symptomatic intracranial haemorrhage was uncommon and did not offset the net benefit in reducing disability. The study results support consideration of endovascular thrombectomy in patients with older age, higher stroke severity, and dominant and non-dominant hemisphere stroke, regardless of ASPECTS or perfusion imaging mismatch. These findings support a major change in prevailing practice regarding the use of endovascular thrombectomy in patients with large-core and very-large-core ischaemic stroke.

effect estimates in important subpopulations, based on clinical and radiological characteristics.

Methods

Study design, search strategy, and trial selection

Analysis of Mechanical Thrombectomy for Large Core Ischemic Stroke (ATLAS) was a patient-level meta-analysis to assess the efficacy and safety of endovascular thrombectomy versus medical management in patients with large-core ischaemic stroke up to 24 h after symptom onset. A research plan was developed a priori and registered with PROSPERO (CRD420251058584). The research protocol, including the primary inclusion criteria for the meta-analysis, is provided in the appendix (pp 2–17). In brief, the meta-analysis included randomised trials that enrolled patients with acute ischaemic stroke due to anterior circulation large-vessel occlusion, and specifically enrolled adult patients with a large-core ischaemic stroke based on use of either ASPECTS⁸ of 5 or lower on non-contrast CT or MR diffusion or, alternatively, estimated ischaemic core volume of 50 mL or greater using CT perfusion or MR diffusion within 24 h of the time the patient was last known to be well; and randomly allocated participants to receive endovascular thrombectomy or best medical care. Detailed eligibility criteria used in each trial are provided in the appendix (pp 27–28). A systematic search using the keywords “stroke”, “thrombectomy”, “large core”, “ASPECTS”, and “ischaemic core”, with results limited to human clinical trials published between March 1, 2018, and March 1, 2025, was executed in PubMed and Embase to identify eligible trials. The complete search strategies for PubMed and Embase are provided in the appendix (pp 17, 26). Additional bibliography and ClinicalTrials.gov registry reviews were undertaken to ensure completeness of the literature review. Authors of the identified eligible completed trials agreed to collaborate by sharing individual patient-level data. This meta-analysis was performed and reported according to PRISMA-IPD guidelines.⁹

Data collection and centralised imaging evaluation

A prespecified list of variables was shared with participating trial investigators. Individual trial statisticians shared data with uniform categorisation and stratification for included characteristics to the ATLAS coordinating centre at UH Cleveland Medical Center (Cleveland, OH, USA). The ATLAS statistical team comprised individuals from the University of Melbourne (Melbourne, VIC, Australia) and UH Cleveland Medical Center, who performed quality assessment and data harmonisation before data analysis.

Investigators from the individual trials shared source neuroimaging collected during the conduct of the trial with the ATLAS central imaging core laboratory at the Royal Melbourne Hospital, University of Melbourne (Melbourne, VIC, Australia). Designated reviewers from

the central core laboratory with representatives from the participating trials re-evaluated all individual patient-level imaging to adjudicate ASPECTS on baseline CT or MRI, ischaemic core volume and perfusion mismatch evaluation, and haemorrhagic transformation. A central core read of ASPECTS by neuroradiologists with at least 10 years of experience was obtained for each patient using 3–5 mm axial slices on 40:40 window settings for CT, and appropriate windowing for diffusion-weighted MRI; all readers were masked to treatment group, the source trial, and the ASPECTS assigned by the trial core laboratory. Patients with an ASPECTS discrepancy of more than 1 point, or any discrepancy in the trichotomy of 0–2 versus 3–5 versus 6–10 points, as compared with the original trial core laboratory read, were reviewed by an additional independent rater from the ATLAS core laboratory, who had access to both previous scores. In cases with a 1-point discrepancy in ASPECTS not affecting the trichotomy, the central read was used. All available perfusion images were reprocessed at the central core laboratory using RapidAI (research version; iSchemaView, Menlo Park, CA, USA), with corrections for motion, bolus timing, and input and output functions where applicable. Ischaemic core volume was estimated using relative cerebral blood flow threshold of less than 30% on CT perfusion imaging or an apparent diffusion coefficient threshold of less than 620×10^{-6} mm²/s for MR diffusion. Critically hypoperfused tissue was defined by a time-to-maximum-intensity threshold of more than 6 s. Two prespecified mismatch profile definitions were used to ascertain the presence or absence of perfusion mismatch: a mismatch ratio ≥ 1.2 and mismatch volume ≥ 10 mL;¹⁰ and a mismatch ratio ≥ 1.8 and mismatch volume ≥ 15 mL.¹¹ Haemorrhagic transformation on follow-up imaging was readjudicated centrally using the radiological categories of the Heidelberg Bleeding Classification¹² by two independent raters with more than 30 years of individual clinical experience; this adjudication was used to reclassify symptomatic haemorrhage across all trials. Occlusion location and the extent of angiographic reperfusion after endovascular thrombectomy was reported as recorded by the trial imaging core laboratory.

Outcomes

The distribution of modified Rankin Scale (mRS) scores at 90 days of follow-up was prespecified as the primary efficacy outcome. Prespecified secondary outcomes included functional independence (mRS score 0–2) and independent ambulation (mRS score 0–3). Safety outcomes included symptomatic intracranial haemorrhage, defined as a large parenchymal haematoma occupying more than 30% of the infarcted territory with substantial mass effect (parenchymal haematoma type 2), or subarachnoid haemorrhage, evaluated by the central core laboratory, with an

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associated increase of at least 4 points on the National Institutes of Health Stroke Scale (NIHSS) within 36 h or death (interval defined based on individual trial criteria); death within 90 days of follow-up; and neurological worsening, defined as an increase of at least 4 points on the NIHSS at 24–48 h from presentation. The pooled proportion of successful reperfusion (modified Thrombectomy in Cerebral Ischemia grade $\geq 2b$) was also reported. 1-year outcomes were not uniformly collected in all participating trials and thus will be reported in a dedicated analysis. A separate analysis detailing haemorrhagic transformation grades and their association with functional outcomes is also planned.

Statistical analysis

The proposed analyses and analytical approaches were described in detail in the study protocol and statistical analysis plan (appendix pp 2–17).

Baseline clinical and imaging characteristics were described for patients randomly allocated to receive endovascular thrombectomy versus medical care. Categorical variables were reported using counts and proportions, whereas continuous variables were reported using medians and IQRs.

The primary outcome of mRS score distribution was evaluated using probabilistic index models (PIMs),^{13,14} adjusting for age, NIHSS at presentation, time from last-known-well to randomisation, and ASPECTS. Respective treatment effects were estimated as adjusted generalised odds ratios (aGenORs), where ties were split equally between the groups, with 95% CIs. Adjusted PIM models differ from parametric ordinal logistic regression models in that PIMs estimate the odds of a random patient from the endovascular thrombectomy group having a better mRS outcome than a random medical management patient, given the differences in all adjustment covariates between these two patients. mRS scores of 5 and 6 were merged into one category to avoid counting transition from mRS score 6 to 5 as treatment success. Missing data were imputed using chained equations, under a missing-at-random assumption. 30 datasets were created using imputed data for individual trials, and were used to generate estimates for PIM models, which were then pooled using Rubin's rule¹⁵ to obtain trial-level estimates. The log-transformed trial-level estimates were pooled through a two-stage random-effects meta-analysis framework using restricted maximum likelihood estimator (REML) random-effects models to obtain overall effect estimates, which were then back exponentiated to provide final aGenOR estimates with 95% CIs. As sensitivity analyses, we also reported results for the best-case scenario for missing 90-day mRS scores (all patients with missing data in the endovascular thrombectomy group assumed to have an mRS of 0, and those in the medical management group assumed to have an mRS of 6) and the worst-case scenario (all patients with missing data in the endovascular

thrombectomy group assumed to have an mRS of 6, and those in the medical management group assumed to have an mRS of 0).

Analyses of all dichotomous secondary outcomes were conducted using a two-stage random-effects meta-analysis of individual participant data. Within-trial estimates were obtained using a modified Poisson regression model with robust standard errors, adjusted for age, NIHSS score at presentation, time from last-known-well to randomisation, and ASPECTS, using the imputed mRS distribution specified above (if applicable). These estimates were pooled using an REML random-effects model to obtain the overall effect estimate, presented as adjusted relative risk (aRR) with 95% CI. Estimates for number-needed-to-treat (NNT) to improve the outcome in one individual were calculated using the following formulas: $1 + (2 / (\text{GenOR} - 1))$, for mRS distribution; and $1 / (\text{pooled standardised risk difference estimate})$, for dichotomous variables. For symptomatic intracranial haemorrhage, a small number of observations precluded adjustment for clinical or imaging covariates. Risk difference estimates for symptomatic intracranial haemorrhage between treatment groups with respective 95% CIs were generated for each individual trial, and pooled using REML.

The prespecified subgroup analyses included age (<70 years vs ≥ 70 years), sex (male vs female), occlusion location (internal carotid artery vs middle cerebral artery), NIHSS score (<20 vs ≥ 20), time from last-known-well to randomisation (dichotomised using two different thresholds, 6 h and 12 h), hemisphere involvement (left vs right), ASPECTS (using categories of 0–2, 3, 4, 5, and 6–10), ischaemic core volume (using categories of <70, 70–100, 100–150, and ≥ 150 mL), and presence or absence of perfusion mismatch using the two prespecified imaging profiles. All analyses were adjusted for age, NIHSS score at presentation, time from last-known-well to randomisation, and ASPECTS on baseline imaging, but each individual adjustment covariate was not included in the subgroup analysis that was based on that covariate (eg, analysis of age-based subgroups was not adjusted for age; appendix pp 23–25). Adjusted PIM subgroup analyses were conducted for the ordinal outcome. Prespecified subgroup analyses were conducted by first estimating trial-level treatment-by-subgroup interaction for individual trials and then pooling resultant estimates using an REML random-effects model.^{16–18} For the analysis of early versus late time window treatment effects within subgroups, formal interaction testing was not done due to the exploratory nature of the analyses and sample size restrictions.

All analyses were done based on treatment assignment at randomisation into individual trials (intention-to-treat), with sensitivity analyses done based on the type of treatment received (as treated). Analyses were done using STATA17 and R version 4.5.1. All hypotheses were

evaluated using two-sided tests and a p value of <0.05 was considered statistically significant. Because there was no prespecified plan for the adjustment of the widths of CIs for the secondary outcomes or for multiple comparisons in the subgroup analyses, the 95% CIs should not be used for hypothesis testing.

Risk of bias

The Cochrane Risk of Bias 2 (RoB 2) tool¹⁹ was used to evaluate the risk of bias across the included trials. The tool assesses the potential risk of bias in different components of clinical trial design and execution. Two assessors (TM and MJ) independently assessed the risk of bias for all participating trials. A third assessor (AS) provided arbitration in case there was no agreement on any individual question. The final assessment was determined based on agreed upon answers in individual domains.

Role of the funding source

There was no funding source for this study.

Results

Six trials met the eligibility criteria and were included in the meta-analysis for all primary, secondary, and safety outcomes. A PRISMA-IPD flow diagram describing the eligible trial selection process and reasons for exclusion in detail is provided in the appendix (p 36). Overall, risk of bias across participating trials was estimated to be low, with some concerns for the secondary outcome of neurological worsening due to missingness of the data in two trials (appendix pp 31–33).

The pooled sample across six trials resulted in 1886 participants: 944 (50.1%) randomly allocated to receive endovascular thrombectomy and 942 (49.9%) to receive medical management. Trial-level population characteristics are provided in the appendix (pp 29–30). Baseline clinical and imaging characteristics for the pooled sample of included patients are provided in table 1. The two treatment groups had a similar median age (70 years [IQR 61–77] in the endovascular thrombectomy group vs 70 years [61–78] in the medical management group), NIHSS score (19 [15–23] vs 18 [15–22]), and time interval from when the patient was last known to be well to randomisation (361 min [220–701] vs 356 min [222–743]). Median values for core laboratory-adjudicated ASPECTS were 4 (3–5) for both groups. Ischaemic core volumes were available for 1520 participants, with median estimates of 79.8 mL (IQR 46.7–127) for the endovascular thrombectomy group (n=759) and 83.0 mL (50.5–138) for the medical management group (n=761). Of 1058 (56.1%) patients with available perfusion mismatch status, 144 (13.6%) did not meet the criteria of mismatch volume ≥ 10 mL with mismatch ratio ≥ 1.2 , and 401 (37.9%) did not meet the criteria of mismatch volume ≥ 15 mL with mismatch ratio ≥ 1.8 . Endovascular thrombectomy was done in

932 patients (926 who were randomly allocated to the endovascular thrombectomy group, and six who crossed over from the medical management group), with substantial reperfusion to more than 50% of the affected arterial territory at the end of endovascular thrombectomy (Expanded Thrombolysis In Cerebral Infarction

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	Endovascular thrombectomy group (n=944)	Medical management group (n=942)	Overall (n=1886)
Trial			
ANGEL ASPECT	230 (24.4%)	225 (23.9%)	455 (24.1%)
LASTE	159 (16.8%)	165 (17.5%)	324 (17.2%)
RESCUE Japan-LIMIT	100 (10.6%)	102 (10.8%)	202 (10.7%)
SELECT2	178 (18.9%)	174 (18.5%)	352 (18.7%)
TENSION	125 (13.2%)	128 (13.6%)	253 (13.4%)
TESLA	152 (16.1%)	148 (15.7%)	300 (15.9%)
Median age, years	70 (61–77)	70 (61–78)	70 (61–77)
Sex			
Male	524 (55.5%)	536 (56.9%)	1060 (56.2%)
Female	420 (44.5%)	406 (43.1%)	826 (43.8%)
Occlusion location			
Internal carotid artery	355 (37.6%)	341 (36.2%)	696 (36.9%)
Middle cerebral artery	587 (62.2%)	601 (63.8%)	1188 (63.0%)
Other	2 (0.2%)	0	2 (0.1%)
Affected hemisphere			
Left	433/944 (45.9%)	431/942 (45.8%)	864/1886 (45.8%)
Right	511/944 (54.1%)	511/942 (54.2%)	1022/1886 (54.2%)
NIHSS score*	19 (15–23)	18 (15–22)	19 (15–22)
Time from last-known-well to randomisation, min	361 (220–701)	356 (222–742)	360 (221–719)
Intravenous thrombolysis given	264/944 (28.0%)	254/941 (27.0%)	518/1885 (27.5%)
ASPECTS†			
Median	4 (3–5)	4 (3–5)	4 (3–5)
0–2	170/944 (18.0%)	188/941 (20.0%)	358/1885 (19.0%)
3	230/944 (24.4%)	210/941 (22.3%)	440/1885 (23.3%)
4	201/944 (21.3%)	220/941 (23.4%)	421/1885 (22.3%)
5	288/944 (30.5%)	265/941 (28.2%)	553/1885 (29.3%)
6–10	55/944 (5.8%)	58/941 (6.2%)	113/1885 (6.0%)
Ischaemic core volume, mL‡	79.8 (46.7–127)	83.0 (50.5–138)	82.1 (48.2–131)
Perfusion mismatch§			
Volume ≥ 10 mL, ratio ≥ 1.2			
No	70/547 (12.8%)	74/511 (14.5%)	144/1058 (13.6%)
Yes	477/547 (87.2%)	437/511 (85.5%)	914/1058 (86.4%)
Volume ≥ 15 mL, ratio ≥ 1.8			
No	197/547 (36.0%)	204/511 (39.9%)	401/1058 (37.9%)
Yes	350/547 (64.0%)	307/511 (60.1%)	657/1058 (62.1%)

Data are n (%), median (IQR), or n/N (%). NIHSS=National Institutes of Health Stroke Scale. ASPECTS=Alberta Stroke Protocol Early CT Score. *NIHSS score at presentation was not available for three participants (one in the endovascular thrombectomy group and two in the medical management group). †ASPECTS at presentation was not available for one participant. ‡Ischaemic core volume was not available for 366 participants (185 in the endovascular thrombectomy group and 181 in the medical management group). §Perfusion mismatch was not available for 828 participants.

Table 1: Baseline clinical and imaging characteristics stratified based on randomisation allocation

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Scale 2b–3) in 771 (82.7% [95% CI 80.2–85.0]) of 932 patients pooled across trials.

In 1871 patients with available mRS scores at the 90-day follow-up, the median mRS score was 4 (3–6) in patients in the endovascular thrombectomy group (n=940) and 5 (4–6) in the medical management group (n=931; figure 1; distributions from the individual trials are provided in the appendix [pp 37–39]). Endovascular thrombectomy improved the mRS score distribution at the 90-day follow-up compared with medical management (aGenOR 1.63 [95% CI 1.42–1.88], $p<0.0001$; ordinal NNT 4.2 [3.3–5.8]; table 2; appendix p 40). At the 90-day follow-up, endovascular thrombectomy also increased functional independence (mRS score 0–2; 183 [19.5%] of 940 patients) compared with the medical management group (70 [7.5%] of 931; aRR 2.57 [95% CI 1.99–3.31], $p<0.0001$; NNT 8.6 [6.2–14.1]; appendix p 41), as well as increasing independent ambulation (mRS score 0–3; 343 [36.5%] vs 185 [19.9%]; aRR 1.96 [1.58–2.43], $p<0.0001$; NNT 5.7 [4.7–7.2]; appendix p 42) and lowering all-cause mortality (292 [31.1%] vs 347 [37.3%]; aRR 0.82 [0.70–0.97], $p=0.022$; NNT 14.6 [7.5–326.3]; table 2).

Symptomatic intracranial haemorrhage, based on central core laboratory assessment of haemorrhagic transformation, was observed in ten (1.1%) of 944 patients in the endovascular thrombectomy group and nine (1.0%) of 942 in the medical management group (pooled unadjusted risk difference –0.17 percentage points [95% CI –1.01 to 0.67 percentage points], $p=0.69$). Among patients assigned to endovascular thrombectomy, symptomatic intracranial haemorrhage occurred in seven (2.7%) of 264 patients who received intravenous thrombolytic medication and three (0.4%) of 680 patients who did not receive intravenous thrombolytic medication. Neurological worsening within 24–48 h occurred in 197 (22.0%) of 896 patients in the endovascular thrombectomy group and 161 (17.9%) of 899 in the medical management group with available data (aRR 1.19 [0.87–1.62], $p=0.27$). Results using best-case scenario (all missing 90-day mRS assumed to be 0 for endovascular thrombectomy group and 6 for medical management group) and worst-case scenario (all missing 90-day mRS assumed to be 6 for endovascular thrombectomy group and 0 for medical management group) were similar to effect estimates for primary outcomes using multiple imputation (appendix p 34). Sensitivity analyses using an

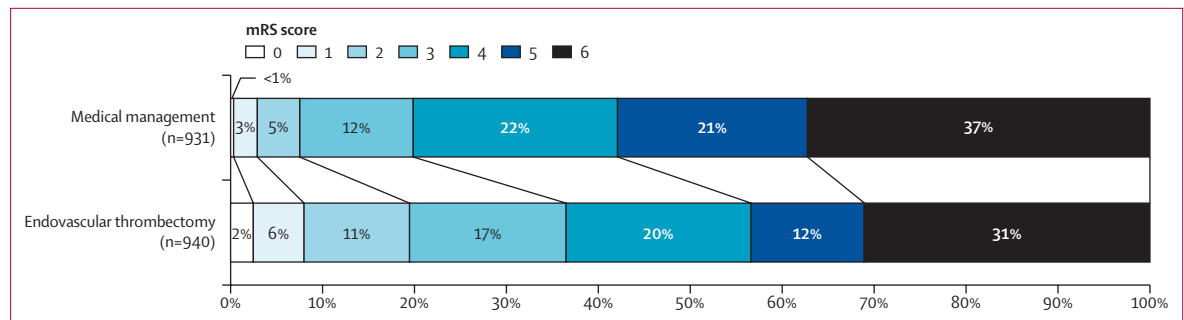


Figure 1: Distribution of mRS scores at 90-day follow-up
mRS=modified Rankin Scale.

	Endovascular thrombectomy group (n=944)	Medical management group (n=942)	Effect estimate (95% CI)	p value	Between-trial heterogeneity		
					τ^2	I^2	Cochran's Q
Primary outcome							
Distribution of mRS scores at 90-day follow-up*	4 (3–6)	5 (4–6)	aGenOR 1.63 (1.42 to 1.88)	<0.0001	0.016	50.4%	10.1, p=0.073
Secondary outcomes							
Functional independence (mRS 0–2) at 90-day follow-up	183/940 (19.5%)	70/931 (7.5%)	aRR 2.57 (1.99 to 3.31)	<0.0001	<0.001	0.0%	5.4, p=0.37
Independent ambulation (mRS 0–3) at 90-day follow-up	343/940 (36.5%)	185/931 (19.9%)	aRR 1.96 (1.58 to 2.43)	<0.0001	0.002	62.0%	13.1, p=0.023
Safety outcomes							
All-cause mortality within 90-day follow-up period	292/940 (31.1%)	347/931 (37.3%)	aRR 0.82 (0.70 to 0.97)	0.022	0.033	48.8%	9.7, p=0.084
Symptomatic intracranial haemorrhage within 36 h	10/944 (1.1%)	9/942 (1.0%)	RD –0.17 (–1.01 to 0.67)†	0.69	<0.001	0.0%	3.3, p=0.66
Neurological worsening within 24–48 h	197/896 (22.0%)	161/899 (17.9%)	aRR 1.19 (0.87 to 1.62)	0.27	0.095	63.9%	13.3, p=0.021

Data per group are median (IQR) or n/N (%) of patients. mRS=modified Rankin Scale. aGenOR=adjusted generalised odds ratio. aRR=adjusted relative risk. RD=risk difference. *Scores were available for 940 patients in the endovascular thrombectomy group and 931 in the medical management group. †Percentage points difference.

Table 2: Functional and safety outcomes in the study population

as-treated approach (appendix p 35) were similar to the primary results reported under the intention-to-treat approach for all primary, secondary, and safety outcomes.

Treatment effect estimates for 90-day mRS ordinal shift within prespecified patient subgroups based on clinical and imaging characteristics are provided in figure 2, with sensitivity analyses presented in the appendix (pp 70–71).

Endovascular thrombectomy consistently improved functional outcomes with no evidence of treatment effect variability in the evaluated clinical and imaging subgroups, except for those with an ischaemic core volume of 150 mL or greater.

Across ASPECTS strata, the treatment effect of endovascular thrombectomy was maintained for

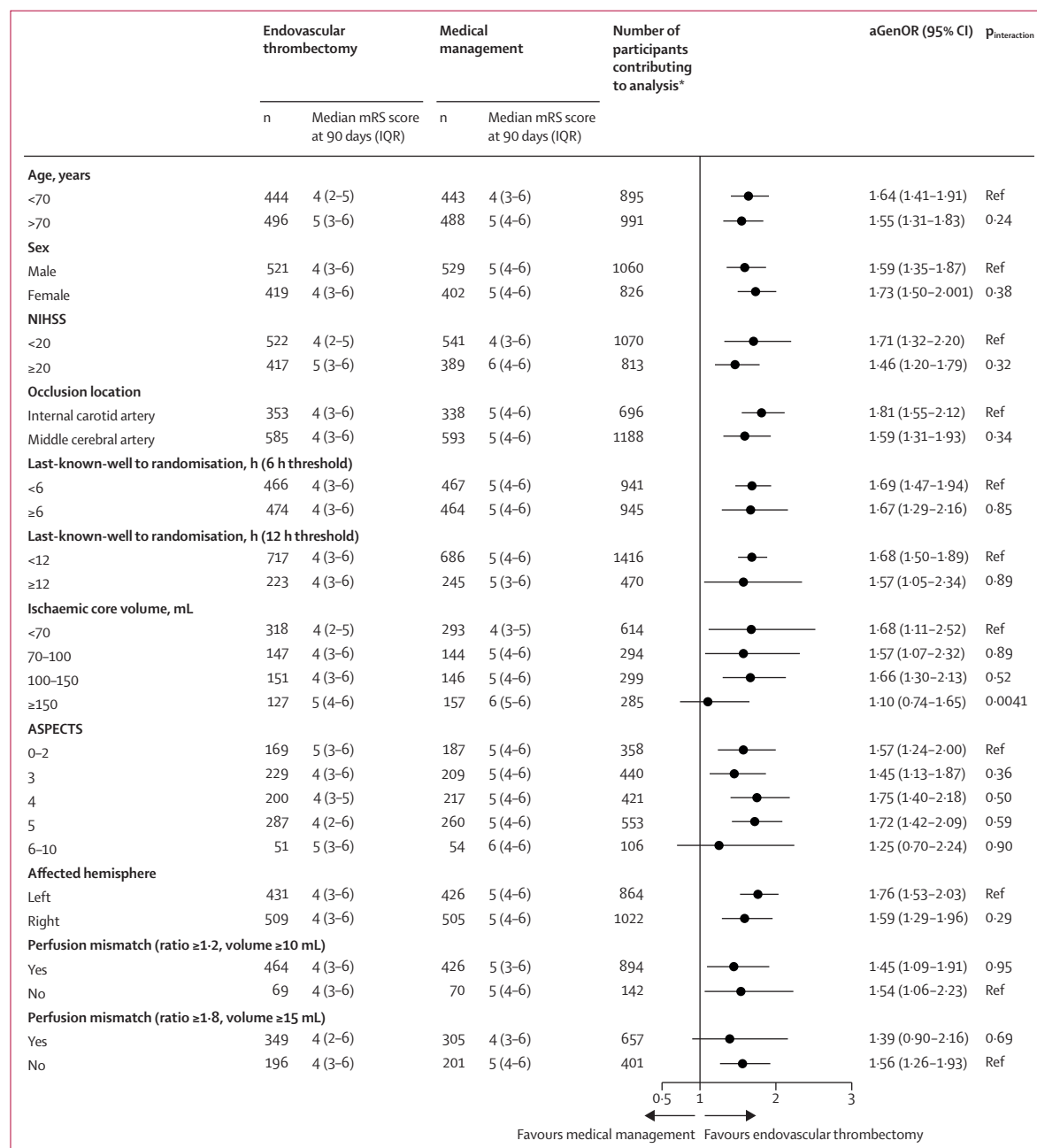


Figure 2: Forest plot of treatment effect estimates across clinical and imaging subgroups

aGenOR=adjusted generalised odds ratio. ASPECTS=Alberta Stroke Protocol Early CT Score. mRS=modified Rankin Scale. NIHSS=National Institutes of Health Stroke Scale. *Missing mRS scores at the 90-day follow-up (n=15) were handled by multiple imputation; thus, the number of patients contributing to the analysis of treatment effect is higher than the number contributing to the median (IQR) values.

ASPECTS 0–2 (aGenOR 1.57 [95% CI 1.24–2.00], n=358, reference group for interaction), 3 (aGenOR 1.45 [1.13–1.87], n=440, $p_{\text{interaction}}=0.36$), 4 (aGenOR 1.75 [1.40–2.18], n=421, $p_{\text{interaction}}=0.50$), and 5 (aGenOR 1.72 [1.42–2.09], n=553, $p_{\text{interaction}}=0.59$). Among patients with ASPECTS 6–10, although treatment effect point estimates favoured endovascular thrombectomy, 95% CIs were wider and overlapped with the null value (aGenOR 1.25 [0.70–2.24], n=106, $p_{\text{interaction}}=0.90$). The magnitude of benefit for patients with ASPECTS 0–2 was substantial, with an NNT of 4.5 (3.0–9.5) with endovascular thrombectomy for improvement of at least 1 mRS point.

Treatment effect estimates favoured endovascular thrombectomy across the ischaemic core volume strata up to 150 mL (figure 2). Specifically, there was benefit in the 100–150 mL subgroup (n=299; aGenOR 1.66 [95% CI 1.30–2.13]; $p<0.0001$; ordinal NNT 4.0 [2.8–7.7]). A significant variation in treatment effect estimates was observed for patients with an ischaemic core volume of 150 mL or greater ($p_{\text{interaction}}=0.0041$ vs those with ischaemic core volume <70 mL); although the point estimate remained favourable, 95% CIs were wide (figure 2). In a sensitivity analysis excluding trials with fewer than 20 patients or fewer than five per treatment group contributing to the subgroup, the results suggested a benefit of endovascular thrombectomy in patients with 150 mL or greater ischaemic core volume (appendix p 71).

Within the subgroups of patients with ASPECTS 0–2, the benefit of endovascular thrombectomy was maintained across the 0–6 h and 6–24 h time windows from last-known-well to randomisation. Benefit was also present in patients with an ischaemic core volume of 100–150 mL in the 0–6 h and 6–24 h time windows. However, in patients with an ischaemic core volume of 150 mL or greater, some trials had insufficient participants for analysis. In subsequent sensitivity analyses excluding trials contributing fewer than 20 patients or fewer than five patients per treatment group in a given subgroup, estimates for the 150 mL or greater subgroup became favourable in the early time window (aGenOR 1.66 [95% CI 1.24–2.22]; appendix pp 72–77).

The treatment effect estimates for the ordinal distribution of 90-day mRS also favoured endovascular thrombectomy among patients with or without perfusion mismatch based on the criteria of mismatch volume ≥ 10 mL and ratio ≥ 1.2 (aGenOR 1.45 [95% CI 1.09–1.91] with mismatch [n=894]; aGenOR 1.54 [95% CI 1.06–2.23] without mismatch [n=142]; $p_{\text{interaction}}=0.95$) and based on mismatch volume ≥ 15 mL with ratio ≥ 1.8 (aGenOR 1.39 [95% CI 0.90–2.16] with mismatch [n=657]; aGenOR 1.56 [95% CI 1.26–1.93] without mismatch [n=401]; $p_{\text{interaction}}=0.69$).

In a post-hoc analysis, treatment effect estimates for the primary outcome at the 90-day follow-up also favoured endovascular thrombectomy in patients aged 80 years or older (aGenOR 1.38 [95% CI 0.99–1.93],

n=326), without evidence of treatment effect variability ($p_{\text{interaction}}=0.19$) compared with patients younger than 80 years (aGenOR 1.64 [95% CI 1.47–1.84], n=1545).

Discussion

This individual patient data meta-analysis of six randomised clinical trials^{2–7} shows strong evidence of the efficacy and safety of endovascular thrombectomy in patients with extensive ischaemic injury up to 24 h after the onset of symptoms. The NNT to improve at least 1 point on the mRS was 4.2 (95% CI 3.3–5.8), compared with an NNT of 2.6 reported in predominantly small-core thrombectomy trials in the early time window.¹ Additionally, patients in the endovascular thrombectomy group had improved functional independence (NNT 8.6 [6.2–14.1]), improved independent ambulation (5.7 [4.7–7.2]), and lower all-cause mortality (14.6 [7.5–326.3]) at 90 days of follow-up compared with those receiving medical management. Early neurological deterioration was slightly more frequent with endovascular thrombectomy (affecting 22.0% of patients) than with medical management (17.9%; aRR 1.19 [0.87–1.62]), but rarely due to symptomatic intracranial haemorrhage, and this did not offset the net clinical benefit. Subgroup analyses showed largely consistent benefit of endovascular thrombectomy across important clinical and imaging characteristics, except for patients with an ischaemic core volume of 150 mL or greater, in whom wide CIs and variable sensitivity analysis results mean that the possibility of a benefit in this group could not be ruled out.

Overall, improved outcomes with endovascular thrombectomy were consistent across the spectrum of age and stroke severity, without any evidence of effect modification. Due to the systematic exclusion of patients older than 80 years or older than 85 years in the included trials (appendix pp 27–28), the proportion of patients aged 80 years or older in this meta-analysis was low (328 [17.4%] of 1886), but similar to those in the HERMES meta-analysis¹ (198 [15.5%] of 1278 patients) and AURORA meta-analysis²⁰ (120 [23.8%] of 504). There was no evidence of treatment effect variability between patients aged 80 years and older versus younger patients. Similar to earlier meta-analyses,¹ the site of arterial occlusion and affected hemisphere did not influence the benefit of endovascular thrombectomy.

The ATLAS meta-analysis substantially expands our understanding of endovascular thrombectomy for patients with ischaemic stroke with a large ischaemic core, including 902 patients with ischaemic core of 70 mL or greater, 607 patients with an ischaemic core of 100 mL or greater, and 297 patients with an ischaemic core of 150 mL or greater. In the HERMES¹ meta-analysis, the numbers of included patients with an ischaemic core of 70 mL or greater were 50 based on CT perfusion and 59 based on MRI,²¹ and the DAWN and DEFUSE 3 trials that pioneered endovascular thrombectomy beyond 6 h after

stroke onset reported median ischaemic core volumes of 8–9 mL.^{11,22} Patients with estimated ischaemic core volumes of 70–100 mL and 100–150 mL had significantly improved outcomes with endovascular thrombectomy in the current analysis based on the primary outcome, suggesting that the overall benefit was not driven by patients with low ASPECTS and smaller ischaemic cores.

The semiquantitative extent of ischaemic change assessed using ASPECTS by the centralised core laboratory did not modify the endovascular thrombectomy treatment effect. Endovascular thrombectomy was associated with improved functional status at 90 days of follow-up among patients with central core laboratory-adjudicated individual ASPECTS of 3, 4, and 5, along with those with very low (0–2) ASPECTS, without evidence of treatment effect variability. Treatment benefit in patients with an ASPECTS of 6–10 was well established by previous trials and the small number of patients in these large core trials led to wide 95% CIs. Of note, while only LASTE⁶ allowed per-protocol inclusion of ASPECTS as low as 0 and contributed 56% of patients with ASPECTS of 0–2 in the early window, 162 patients had ASPECTS of 0–2 in the late window in this pooled data.

Prespecified analyses of the effect of endovascular thrombectomy based on the primary outcome in the 0–6 h and 6–24 h time windows were done within the subgroups of patients with the most extensive ischaemic cores (patients with ASPECTS of 0–2 and estimated ischaemic core ≥ 150 mL). Sensitivity analyses were done to exclude trials with limited representation of these subgroups to ascertain the stability of treatment effect estimates. Among patients with ASPECTS of 0–2, treatment effect estimates favoured endovascular thrombectomy across both early and late time windows with no significant variability. While current stroke guidelines from the American Heart Association and American Stroke Association²³ designate endovascular thrombectomy as reasonable for patients with ASPECTS of 0–2, our meta-analysis provides evidence that clinical benefit is not merely possible, but highly probable in the early time window and likely also the late time window. In patients with an ischaemic core volume of 150 mL or greater, the treatment effect of endovascular thrombectomy had a favourable point estimate with a wide 95% CI. However, sensitivity analyses suggested a benefit, particularly in the early time window, and so we cannot exclude a benefit of endovascular thrombectomy in this patient group.

Perfusion mismatch evaluation was available in 1058 (56.1%) of 1886 participants, and a substantial number of patients did not have mismatch in these trials. Treatment effect estimates based on the primary outcome favoured endovascular thrombectomy consistently among patients with and without mismatch based on the prespecified definitions, without evidence of treatment effect variability. Previous results have shown that larger penumbra size might correlate with the magnitude of treatment benefit.²⁴ While we did not prespecify absolute

penumbral volume as an outcome, future dedicated analyses have been planned to examine penumbral characteristics in more depth. Of note, none of the trials set upper limits for ischaemic core volume or required perfusion mismatch for eligibility in patients with ASPECTS of 3–5, and only patients with an estimated ischaemic core volume of 100 mL or greater were excluded if CT-assessed ASPECTS was 0–2 in the ANGEL ASPECT trial.³

The question of whether imaging beyond non-contrast CT or diffusion-weighted MRI is needed beyond 6 h after onset is important. None of the trials required perfusion mismatch for eligibility (within or beyond 6 h after stroke onset) and our data showed no evidence of variability in treatment effect based on the presence or absence of perfusion mismatch. Therefore, in patients who meet the clinical eligibility for these trials, we do not have any evidence to recommend further patient selection using advanced imaging techniques or parameters beyond excluding oedematous subacute infarction.

Salvage of ischaemic penumbra through reperfusion has been shown to reduce disability in ischaemic stroke. However, the traditional dichotomous concept of salvageable versus irreversibly injured tissue is likely to be an oversimplification.²⁵ Ischaemic injury thresholds vary, with grey matter being more susceptible to ischaemic injury than white matter and other supporting tissue, potentially resulting in an injury gradient.^{26,27} Effective reperfusion with endovascular thrombectomy could prevent pan-necrosis, supporting the hypothesis of partial tissue salvage within brain areas that were previously thought to be irreversibly injured (including hypodensity on non-contrast CT and restricted diffusion on MRI). These mechanisms also explain, at least in part, the benefit observed with endovascular thrombectomy and reperfusion in patients without perfusion mismatch. The traditional definitions of mismatch requiring a ratio of ≥ 1.2 or ≥ 1.8 ,^{10,11} developed in patients with relatively small-core stroke, might also not be appropriate for large-core stroke (eg, a patient with a 100 mL core volume would need >80 mL mismatch to meet a ratio of >1.8 , but far less tissue salvage is likely to be required to reduce post-stroke disability).

While slightly more frequent in the endovascular thrombectomy group than the medical management group, neurological worsening in our analysis did not reach statistical significance. Importantly, symptomatic intracranial haemorrhage occurred in only 1.1% of patients in patients randomly allocated to receive endovascular thrombectomy, as readjudicated by the central core laboratory, and was less frequent than reported in previous endovascular thrombectomy trials, with no difference observed between the endovascular thrombectomy and medical management groups. Most neurological deterioration was associated with cerebral oedema rather than symptomatic intracranial

haemorrhage. Importantly, the trials included in this meta-analysis excluded patients with established oedema with mass effect, and the results thus do not apply to those patients.

While endovascular thrombectomy in the early or late time windows did not reduce mortality in patients with small-core stroke in the HERMES and AURORA meta-analyses,^{1,20} there was a significant reduction in mortality in a study-level meta-analysis that combined early and late window trials, with an NNT to prevent one death of 31.²⁸ In the current meta-analysis of trials enrolling patients with severe ischaemic stroke, the NNT to prevent one death was 14·6 (95% CI 7·5–326·3). Furthermore, although severe disability requiring constant care (mRS score 5) was not a formal outcome measure, endovascular thrombectomy appeared to reduce the proportion of patients in this category (figure 1). By shifting the outcome distribution away from mortality and near-complete disability, endovascular thrombectomy provides valuable quality-of-life benefits—and likely economic benefits—in those with large-core ischaemic stroke.²⁹ Consequently, these findings justify wider acceptance of thrombectomy in this population from a societal point of view, despite an overall lower proportion of patients with large-core stroke having functional independence and independent ambulation compared with patients with small-core stroke.

Even with consistent benefit of endovascular thrombectomy in this population, the disability reduction remains suboptimal: one in three patients have independent ambulation (mRS 0–3) and only one in five have functional independence (mRS 0–2) at 3 months. Therefore, there is substantial need and opportunity to further optimise clinical outcomes in this population. These patients with more evolved ischaemia might also benefit from strategies targeting improved microvascular tissue reperfusion,^{30,31} reduced cerebral oedema, and use of brain cytoprotective agents.

This meta-analysis showed a strong treatment benefit of endovascular thrombectomy in patients with large-core ischaemic stroke across an geographically diverse population and a wide range of clinical and imaging characteristics. These results, in conjunction with individual trials, support endovascular thrombectomy as the standard of care for most patients with large-core ischaemic stroke.^{2–7,32–34} The absence of treatment effect modification by most imaging characteristics in this meta-analysis indicates that the outer limits of endovascular thrombectomy benefit are unlikely to be represented by one specific threshold for ischaemic core or ASPECTS when considering who to treat with thrombectomy. Individual trials have reported imaging characteristics that might modify endovascular thrombectomy treatment benefit—including severe hypodensity or net water uptake on non-contrast CT at presentation^{35,36}—which, if further validated, could become useful selection approaches in the clinic. Importantly, the trials enrolled patients with early ischaemic changes, and the results do not apply to patients

with established large infarcts, cerebral oedema, and mass effect. Similarly, patients with pre-existing disability were not included. The likely prognosis needs to be considered using a synthesis of imaging profiles, the frailty and comorbidity of the patient, and their personal attitudes to disability tolerance.³⁷ Such decisions are challenging in a time-pressured environment and, in many cases, it might be appropriate to default to treat the patient and defer prognostication to the subsequent days when the outlook is usually clearer.

This meta-analysis has some limitations. From a methodological standpoint, some of the included trials excluded certain subgroups such as patients older than 80 years or with ASPECTS of 0–2. However, the overall individual patient-level data analysis included more than 300 patients older than 80 years and more than 300 with ASPECTS of 0–2, providing a large sample to analyse. The individual trials also excluded patients with established ischaemic changes and significant cerebral oedema, warranting caution in interpreting the results in these patients with advanced ischaemia. We dichotomised continuous characteristics to report subgroup analyses based on clinical relevance and existing endovascular thrombectomy literature,^{2–7} but this approach might have led to the loss of some information about treatment effect variability compared with continuous analysis.

Strengths of the analysis include harmonised imaging parameter evaluation through use of a centralised core laboratory and uniform image processing protocols, and a large number of included participants, resulting in higher precision and increased power for subgroup analyses. Furthermore, the inclusion of diverse populations, demographics, and systems of care contributes to the generalisability of findings and emphasises the practicality of endovascular thrombectomy across systems that use different imaging modalities (CT, MRI, or perfusion imaging) for ischaemic injury assessment and thrombectomy decision making.

In conclusion, this individual patient-data meta-analysis of six randomised controlled trials shows improved functional outcomes and reduced mortality with endovascular thrombectomy in patients with large-core ischaemic stroke at baseline, without an increase in the rate of symptomatic intracranial haemorrhage. Endovascular thrombectomy remained beneficial regardless of age, sex, stroke severity, occlusion location, hemisphere involvement, or perfusion mismatch. With the exception of very extensive ischaemic changes (core volume ≥ 150 mL) presenting in the late time window, where evidence remains limited, benefit was sustained across ASPECTS and ischaemic core strata for patients presenting up to 24 h after onset.

Contributors

Conceptualisation, data curation, funding acquisition, investigation, methodology, project administration, resources, supervision, and writing (original draft): AS, BCVC, SY, TM, ZM, XH, GT, MB, VC, TJ, BL, CA, AJY, and OOO. Formal analysis and visualisation: AS, BCVC, SY, TM,

ZM, XH, GT, MB, VC, TJ, BL, CA, AJY, OOO, LC, HJ, and DP. Writing (review and editing): all authors.

Declaration of interests

AJY declares grants and contracts paid to their institution from Medtronic, Cerenovus, Penumbra, Stryker, Genentech, Balt, and Route 92; consulting fees from Penumbra, Vesalio, and NIH/NINDS; participation on a data safety and monitoring board or advisory board with Route 92 and HCA Neurovascular Research Advisory Board; and stock or stock options in Insera Therapeutics, Galaxy Medical, Gravity Medical, and Nicolab. AEH declares consulting fees from Medtronic, Microvention, Stryker, Penumbra, J&J MedTech, Genentech, GE Healthcare, Scientia, Balt, Viz.ai, Insera therapeutics, Proximie, NeuroVasc, NovaSignal, Vesalio, Rapid Medical, Imperative Care, Galaxy Therapeutics, Route 92, Perfuzze, CorTech, Imago, Shockwave, Toro, NeuroVasX, Xcath, Kaneka, Plaga, Project Neu, Piraeus, BioPhiliQ, Arbor Medical, Medinol, Navigantis, and Magnisity; payment or honoraria for lectures, presentations, speakers bureaus, manuscript writing, or educational events from Medtronic, Microvention, Stryker, Penumbra, J&J MedTech, Genentech, GE Healthcare, Scientia, Balt, Viz.ai, Insera therapeutics, Proximie, NeuroVasc, NovaSignal, Vesalio, Rapid Medical, Imperative Care, Galaxy Therapeutics, Route 92, Perfuzze, CorTech, Imago, Shockwave, Toro, NeuroVasX, Xcath, Kaneka, Plaga, Project Neu, Piraeus, BioPhiliQ, Arbor Medical, Medinol, Navigantis, and Magnisity; participation on a data safety and monitoring board for the COMAND trial and FOCUS trial; and being Past President of the Society of Vascular and Interventional Neurology. AS declares grant support as a global principal investigator (PI) for SELECT LATE (Patient-Centered Outcomes Research Institute; paid to their institution) and as a PI for SELECT, SELECT2, and Stryker Neurovascular (paid to their institution). AR declares consulting fees from Stryker NV, Cerenovus, Route 92; and participation on a data safety and monitoring board for ATHENA and TORNADO. BG declares grant support from the French Ministry of Health as a PI for TITAN, DIRECT ANGIO, and IA-SUCCESS; consulting fees from Air Liquide, MIVI, Medtronic, Microvention, Perfuzze, and Penumbra; payment or honoraria for lectures, presentations, speakers bureaus, manuscript writing or educational events from Boehringer Ingelheim, Acticor Biotech, Medtronic, and Penumbra. BL declares grants or contracts from Balt, Microvention, and Phenox; and consulting fees from Olea Medical. CA declares consulting fees from Medtronic; and payment or honoraria for lectures, presentations, speakers bureaus, manuscript writing or educational events from Medtronic. CM declares grants or contracts from the European Union, CVON/Dutch Heart Foundation, Boehringer Ingelheim, and Stryker (paid to their institution); and stock or stock options from Nicolab (minority interest). CZS declares grants or contracts from the Health Research Foundation of Central Denmark Region (research grant from hospital); and participation on a data safety monitoring board or advisory board for EAST-STROKE. DSL declares consulting fees for imaging core laboratory work from Johnson & Johnson and Rapid Medical. DD declares participation on a data and safety monitoring committee for ACT Global, and, in the past, data and safety monitoring committee for TESLA; and role as Vice Chair of the data access committee of CONTRAST consortium. GM declares consulting fees from Stryker, Terumo Neuro, and Sim & Cure; payment for lectures for Penumbra, Medtronic, Wallaby Phenox, Cerenovus (Johnson & Johnson), and Bracco; and participation on an advisory board for the MAESTRO study. GT declares (related to the current manuscript) receipt of funding for the TENSION trial from the EU Horizon 2020 Research and Innovation Programme under grant agreement number 754640; consulting fees from Acandis, Bayer, Boehringer Ingelheim, and TarGED; payment or honoraria from Acandis, Boehringer Ingelheim, and Lilly; and role as Chair of the Board of Directors of the European Stroke Organisation. HY declares payment or honoraria for lectures, presentations, speakers bureaus, manuscript writing or educational events from Daiichi-Sankyo, Otsuka Pharmaceutical, Stryker, Boston Scientific, Abbott Medical (lecture fee, personal). JCG declares research grants from Genentech (supplies investigational product); consulting fees from Frazer (consulting on mobile stroke unit); payment for expert testimony from various law firms (legal malpractice cases) three times in 2026; support for attending meetings and/or travel from Perosphere (consulting meeting); participation as chair of a data safety monitoring

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Reist/Q'Apel Medical, Truvic, Tulavi Therapeutics, Vastrax, Piraeus Medical, Brain4Care, Quantanosis AI, Unity Medical, and Viseon. SOG reports research grants from NIH/NINDS, PCORI, Stryker, Medtronic, and Methinks; consulting fees from Stryker and Medtronic; payment or honoraria from Stryker and Medtronic; appointment as president of the Society of Vascular and Interventional Neurology; and stock options and equity in Motif, Gravity, and Clotter. TNN reports consulting fees as an advisor for Bayer, Route 92, and Medtronic; and appointment as Associate Editor for *Stroke*. All other authors declare no competing interests.

Data sharing

Individual patient-level data will not be made available. Statistical codes and outputs will be shared for the purpose of replicating results after a reasonable request to the corresponding author is reviewed and approved by the ATLAS collaboration steering committee. The study protocol and statistical analysis plan are included in the appendix.

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